

Lab Assignment – 3

Problem

Implement a TCP client-server application in C using socket programming. The server should accept a connection from a client and provide a simple integer statistics service.

The client reads a set of integers from the user and sends them to the server. The server computes the **minimum, maximum, sum, and average** of the received integers and sends the results back to the client. The client displays the results.

The server should continue accepting new client connections until it is terminated manually.

Inputs

Server:

- Port number on which the server will listen.

Client:

- Server IP address.
- Server port number.
- Number of integers N.
- N integer values.

Output

Server should display:

- Client IP address and port number after a connection is accepted.
- Integers received from the client.
- Computed minimum, maximum, sum, and average.
- A message when the client disconnects.

Client should display:

- Confirmation that connection to the server is established.
- Minimum, maximum, sum, and average received from the server.

Example

Client Input:

N = 5

Values: 12 7 25 4 18

Client Output:

Connected to server.

Minimum = 4
Maximum = 25
Sum = 66
Average = 13.20